


# PRESENTATION TOPIC: HTTP VERSION 1

**Course Title: Data Communication**

**Course Code: ICT 3101**

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# WHAT IS HTTP/1

- HTTP is the **Hypertext Transfer Protocol** of the World Wide Web.
- It is an **application-layer** protocol that runs over **TCP**.
- It follows a **client–server, request–response** model.
- Messages are **plain text** and **stateless**.
- HTTP/1 covers versions **0.9, 1.0 and 1.1**.

# WHY HTTP IS USED

- Fetches **web pages, images, and files** from servers.
- **Simple** and easy to read, write, and debug.
- Works with **any content type** via MIME headers.
- Supported by **every browser, server, and proxy**.
- Foundation of **websites, REST APIs, and web services**.

# HISTORY OF HTTP/1

| | |
|---|---|
| **1991** | HTTP/0.9 by Tim Berners-Lee at CERN — GET only, HTML only. |
| **1996** | HTTP/1.0 (RFC 1945) adds headers, status codes, POST and HEAD. |
| **1997** | HTTP/1.1 (RFC 2068) adds persistent connections and Host header. |
| **2022** | HTTP/1.1 restated in RFC 9110 and RFC 9112. |

# HTTP/1 COMMUNICATION STACK

**WEB BROWSER (CLIENT)**

Chrome, Firefox, curl — sends the request.

**HTTP (APPLICATION LAYER)**

Request and response text messages.

**TCP (TRANSPORT LAYER)**

Reliable connection — port 80 by default.

**IP (NETWORK LAYER)**

Addressing and routing of packets.

**WEB SERVER**

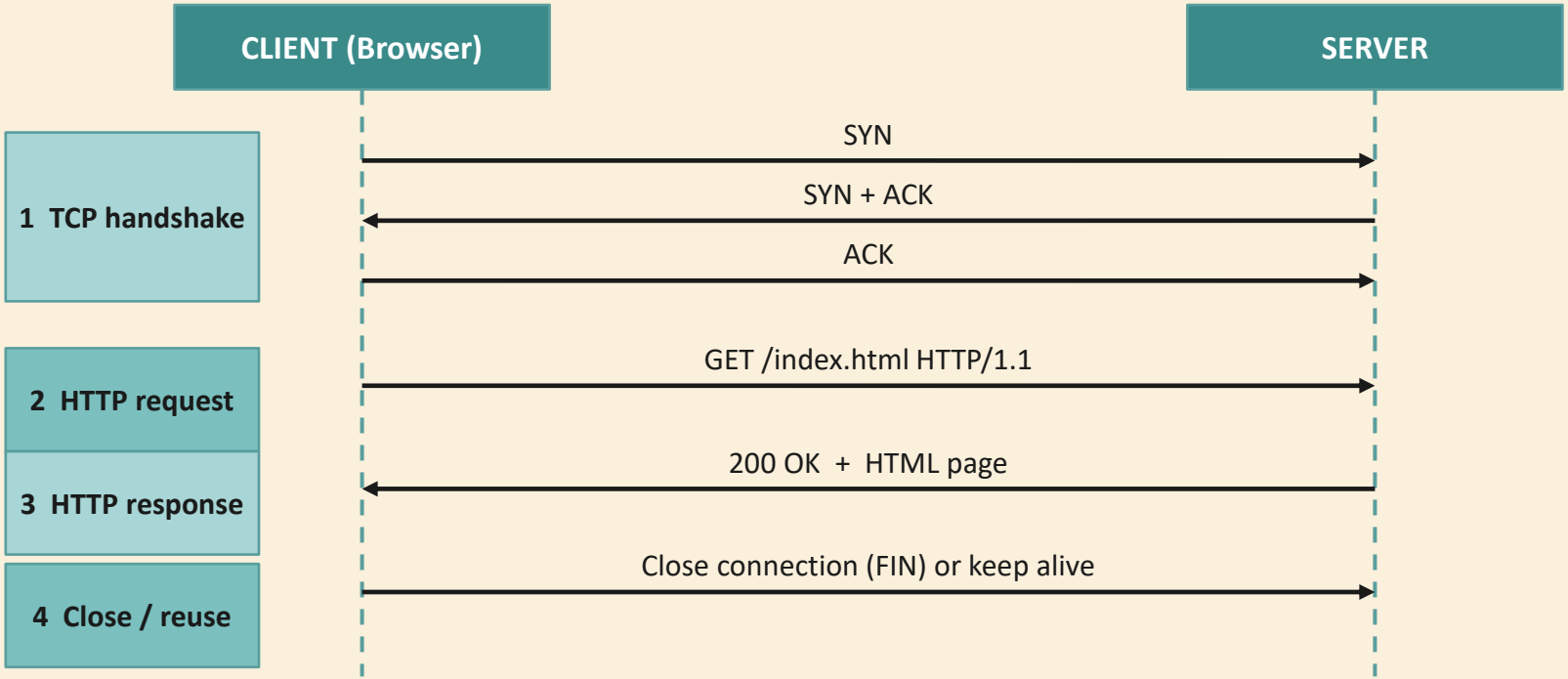
Apache, Nginx, IIS — sends the response.

# HOW HTTP/1 WORKS

## HTTP Request–Response Cycle — Step by Step:

1. DNS Lookup — Browser resolves the domain to an IP address.
2. TCP Handshake — Client opens a connection to the server.
3. HTTP Request — Client sends method, URL, and headers.
4. HTTP Response — Server returns status code, headers, and body.
5. Close or Reuse — Closed in 1.0; kept alive in 1.1.

# HTTP REQUEST–RESPONSE DIAGRAM



# HTTP REQUEST METHODS

**Actions a client can ask the server to perform:**

### GET

Use: Retrieve a resource.

### POST

Use: Submit data to server.

### HEAD

Use: Get headers only.

### PUT

Use: Upload or replace (1.1).

### DELETE

Use: Remove a resource (1.1).

### OPTIONS

Use: List allowed methods (1.1).

# HTTP MESSAGE EXAMPLE

**HTTP REQUEST**

```
GET /index.html HTTP/1.1
Host: www.example.com
User-Agent: Mozilla/5.0
Accept: text/html
```

- Request line (method, URL, version)
- Headers (name: value)
- Blank line ends headers

**HTTP RESPONSE**

```
HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: 1256

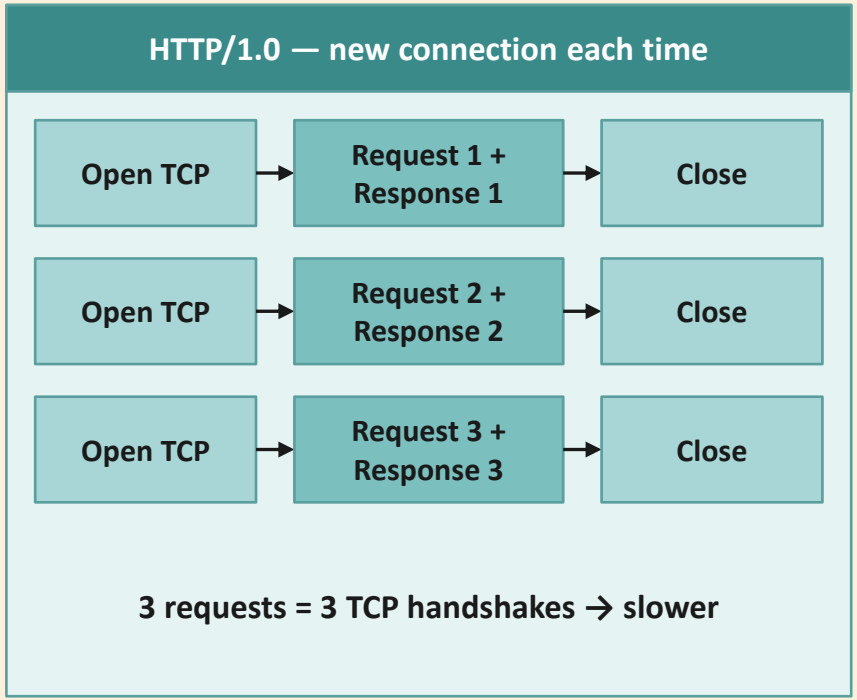
<html> ... </html>
```

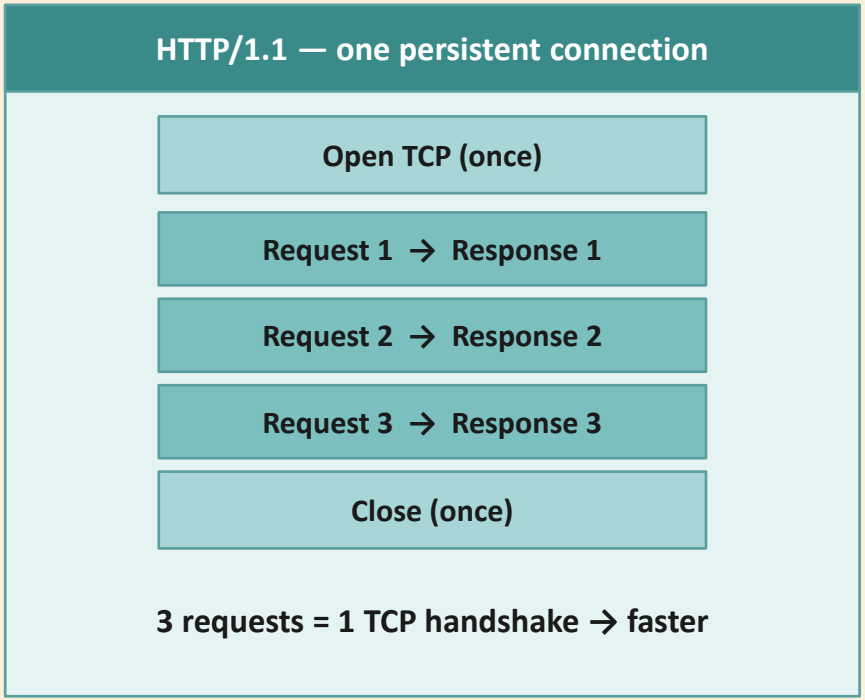
- Status line (version, code, text)
- Headers
- Blank line
- Body (the page content)

# HTTP/0.9 vs 1.0 vs 1.1

| Feature | HTTP/0.9 | HTTP/1.0 | HTTP/1.1 |
|---|---|---|---|
| **Connection** | One per request | One per request | Persistent |
| **Methods** | GET only | GET, HEAD, POST | Adds PUT, DELETE |
| **Headers** | None | Introduced | Host required |
| **Status Codes** | None | Introduced | Expanded |
| **Caching** | None | Basic | Cache-Control, ETag |
| **Content** | HTML only | Any MIME type | Chunked transfer |

# HTTP/1.0 vs HTTP/1.1 CONNECTIONS



# ADVANTAGES OF HTTP/1

- **Simple** and human-readable text format.
- **Persistent connections** reduce delay (1.1).
- **Host header** allows many sites on one IP.
- **Caching** saves bandwidth and time.
- **Universal support** across browsers and servers.

# COMMON STATUS CODES

### 200 OK

Request succeeded; resource returned.

### 301 Moved Permanently

Resource has a new URL; browser redirects.

### 304 Not Modified

Cached copy is still valid.

### 403 Forbidden

Server refuses to allow access.

### 404 Not Found

Requested resource does not exist.

### 500 Server Error

Server failed to handle the request.

# LIMITATIONS OF HTTP/1

- **Head-of-line blocking** on a connection.
- **Many TCP connections** needed for one page.
- **Repeated, uncompressed** headers waste bandwidth.
- **No encryption** — data is readable by others.
- **Stateless** — needs cookies for sessions.
- **No server push** — later fixed by HTTP/2.

# SUMMARY

- HTTP/1 is a **text-based request–response** protocol over **TCP**.
- Versions: **0.9 (1991)**, **1.0 (1996)**, **1.1 (1997)**.
- Core parts: **methods, headers, status codes, and body**.
- HTTP/1.1 added **persistent connections, Host, and caching**.
- Its limits led to **HTTP/2 and HTTP/3**; **HTTPS** adds security.

# THANK YOU

Any Questions Are Welcome